

Depth-Based UV Intensity and Temperature at Dana Point

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Abstract—Climate change, pollutants, and other anthropogenic stressors have impacted global ocean surface temperatures and ultraviolet (UV) exposure levels. Understanding how temperature and UV radiation vary with ocean depth can help convey the changing environmental conditions many marine organisms experience. This project investigates how temperature and UV intensity change with depth in the ocean water at Dana Point Pier, California, using an autonomous underwater vehicle (AUV) to collect data. This paper presents the design, validation and deployment of the AUV at Dana Point Pier, California. The AUV was equipped with three sensors: an MPX5700 pressure sensor to determine depth, a Murata NXFT15WB473FA2B150 thermistor to measure temperature, and finally a GUV-A-S12SD sensor to measure UV intensity. This project's objective was to use pressure, temperature, and UV to analyze how UV intensity and temperature vary with ocean depth. Both quantities are influenced by sunlight, as the sun is the primary source of natural UV light and helps warm the surface of the ocean. The most meaningful result obtained from the deployment at Dana Point was that UV experienced an exponential decay, decreasing approximately 46.2% per meter. Furthermore, temperature remained relatively constant during the descent from 0.5 to 3 meters with an average reading of 18.1 Celsius. The main takeaway from this project is that experimentation can be represented into 4 main categories: calculations, simulations, prototyping, and testing. This was evident in the context of this climate change project.

Keywords—Climate Change, AUV, Pressure, Temperature, and UV

I. INTRODUCTION

The purpose of this project is to successfully design, test, and deploy a fully functional autonomous underwater robot that would successfully dive to a depth of 3 meters to collect meaningful environmental data of nearby aquatic life [1]. This was done through deploying the AUV at Dana Point Dock that measured local temperature and UV at varying depths.

The motivation behind this project was to understand how climate change is and can have an impact on many aquatic ecosystems. In recent research papers, Priya A. K shares, "Stresses caused by climate change, such as rising and hotter seas..., threaten the world's most populated areas and aquatic ecosystems." [6] Thus as a result measuring these qualities will develop a well rounded understanding of how humans impact local aquatic ecosystems.

Ultraviolet was measured as one of the quantities of consideration since many organisms such as some fish rely on

UV light for vision. However, on the contrary many organisms such as plankton can experience lethal harm from UV light exposure. In recent research papers, D-P Hader shares, "Recent results have demonstrated the negative impacts of exposure to UV-B radiation on primary producers, including effects on cyanobacteria, phytoplankton, macroalgae, and aquatic plants." [5] This conveys that disruption to the atmosphere caused by climate change, negatively impacts the ecosystem through the increased rate of UV exposure.

It was expected that UV would decrease with depth water absorbs and scatters light.[8] A model that is useful in approximating this behavior would be, Beer-Lamberts Law which states that light decays exponentially with respect to depth. This can be model with Equation 1:

$$UV = k_0 * \exp(-k * z) \quad (1)$$

Where k_0 represents the intensity of light just below the water's surface, k the diffusion attenuation coefficient, and z the depth.

Temperature was another quantity that was measured through the deployment of the AUV. Temperature is critical to the well being of many aquatic organisms, as the slightest change in temperature can harm many aquatic organisms. In a research paper, the International Fund for Animal Welfare, shares that "However, these fish are incredibly sensitive to temperature and, as waters get warmer, their metabolism quickly increases." [7] This increase in metabolism would mean that the fish would have utilized more oxygen for normal tasks demanding more on their bodies.

Pressure was the last quantity that was measured due to its direct relationship with depth. Through quantifying pressure, depth can be measured which would further the relationship between temperature, UV with depth. However, through leveraging a pressure sensor, a control effort, specifically PID control, was developed that would dynamically adjust the diving robots position in order to reach each desired depth.

With the basic familiarization of the motivation that went behind this project, it is important to understand the tools that went into developing this project. Operational amplifiers (Op-amps) are highly versatile devices that specialize in amplifying and filtering signals. In the case of E80, op amps were utilized to amplify signals as well as to minimize noise. The general equation that explains this relationship is conveyed in Equation 2:

$$V_{out} = (1 + \frac{R_f}{R_{n1}})(\frac{R_g}{R_{p1} + R_g})V_{p1} - (\frac{R_f}{R_{n1}})V_{in} \quad (2)$$

Equation 2 refers to the output voltage read and converted by the Teensy. V_{in} refers to the voltage being supplied in the to circuit, in the case of all the circuits it was a standard 5V. R_f refers to the feedback resistance, which connects the the output to the inverting input. R_{n1} refers to the input resistance. R_g refers to the external resistor connected to ground responsible for offsetting. Finally R_{p1} refers to the initial resistance V_{p1} experiences before the voltage divider in pin 5.

The sensors that were chosen for this project had their voltage range scaled to fit in the acceptable voltage analog range of the Teensy 4.0. The Teensy, a 10 bit microcontroller, was utilized to read in analog voltage data measured by the sensors. The acceptable range of voltages for the Teensy are, 0 - 3.3 V, however to give each sensor a buffer and a larger margin of error, each sensor was scaled and offsetted to have voltage readings be between 0.3 - 3V as seen by Equation 3.

$$Gain = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{R_f}{R_{n1}} \quad (3)$$

For the E80 robot, the sensors-thermistors, pressure, and UV were all scaled to be within the range of 0.3 - 3V, to increase measurement resolution, as well as to avoid the Teensy clipping of any measurements that were near boundary ranges.

First, this paper will communicate about the design choices that went behind designing each sensor. Then, the design choices that went to developing the mechanical robot. Next, the deployment details that took place to result in a successful trip. Finally, the results are analyzed and the conclusion of the project.

II. SENSOR DESIGN

A. Pressure Sensor

The pressure sensor was pivotal for a diving robot design as the control effort applied to the autonomous underwater robot was dependent on the robot's depth. Since pressure is dependent on depth, developing a highly accurate pressure sensor was essential. The integrated silicon pressure sensor, MPX5700, was chosen for its fast response time as well as highly accurate voltage output reading [3]. To develop a sensor that would accurately measure the pressure up to 3 meters deep it was necessary to define the expected input values the Teensy would read in. Since the pressure sensor converts pressure to voltage, the pressure sensor datasheet defines the voltage out from the pressure sensor to be equal to the supply voltage, V_s , multiplied by the absolute pressure, at a depth, as defined by Equation 4. As defined by Equation 4, P_{atm} refers to the atmospheric pressure, ρ refers to the density of liquid, g refers to the gravitational acceleration, and finally h refers to the depth within the liquid.

$$V_{in} = V_s * (0.0012858 * (P_{atm} + \rho * g * h) + 0.04) \quad (4)$$

Through leveraging Equation 4, for depth ranges from 0 - 3 m, an inverting gain amplifier was utilized that mapped 0.3 V to 3 m and 3V to 0.3 m through leveraging Equation 2 and 3

to find the appropriate gain and offset. Demonstrated in Figure 1, is a schematic of the pressure circuit that was designed.

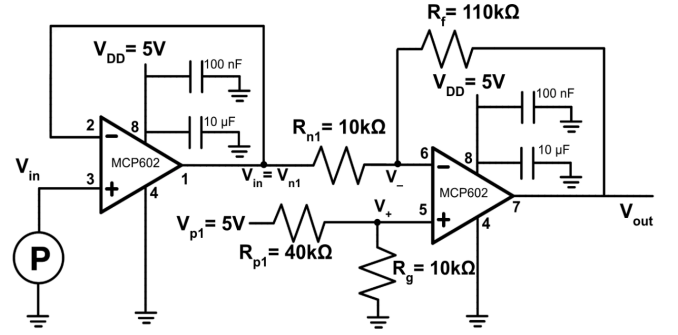


Figure 1: Pressure Sensor Voltage-to-Voltage Interface Circuit, includes Unity Gain Buffer

The sensor had to be calibrated to convert the output voltage to an equivalent depth. To do this a Depth vs. Voltage graph was developed through measuring the output voltage of the pressure sensor to a known depth. To accurately determine the measurements, a meter stick was utilized to measure depths in increments of 0.1 m to get output voltage. To further increase the accuracy of the calibration curve, field measurements at Dana Point were done to generate a calibration curve. Since the density of salt water, like Dana Point, differs compared to fresh water, 1020 kg/m³ compared to 1000 kg/m³ respectively. The calibration curve generated in Dana Point looks as such:

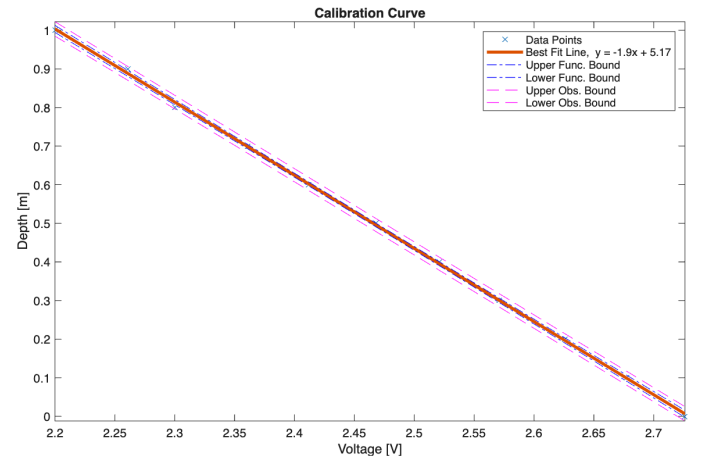


Figure 2: Calibration curve of Depth (m) vs. Voltage (V)

As conveyed in Figure 2, the best fit line generates a line of slope -1.9 and intercept 5.17. These values generated from the calibration curve were used in developing the control effort for the diving robot. Thus with the pressure sensor developed with its respective calibration curve, sensors such as temperature and UV were able to plot with respect to depth of the robot.

B. Thermistor Sensor

The thermistor was an extremely pivotal sensor in developing a relationship between temperature and depth. As conveyed before, a thermistor is a highly sensitive resistor that changes significantly to varying temperatures. The thermistor, Murata NXFT15WB473FA2B150, was chosen due its

accuracy, response time, as well as simplicity [2]. Through interactions with the sensor in Lab 4, the response time of the thermistor was measured to be around 7 seconds, with the sensor accurate up 1 C, and only a voltage divider needed to get a voltage, the thermistor presented itself as the best candidate. In order to develop a circuit that is able to leverage this sensor, expected input value ranges had to be defined. Through research, in recent years, data conveyed that the average surface temperature of Dana Point water around mornings ranged from 10 - 20 C [9]. With the temperature range defined, Equation 5 was leveraged to find the resistance of the thermistor at a given temperature. Where R_o refers to the resistance at room temperature (25 C), 4050 refers to the Beta constant, and finally T refers to the temperature of liquid being measured. To which Equation 6 was utilized to find the expected voltage entering the system. Where R2 was chosen to be the resistance value of the thermistor at room temperature, R_t was the resistance of the thermistor itself, and V_{in} would represent the supply voltage of 5V.

$$R = R_o * \exp(4050 * (\frac{1}{T + 273.15} + \frac{1}{298.15})) \quad (5)$$

$$V_{out} = \frac{R_2}{R_t + R_2} * V_{in} \quad (6)$$

Through having the expected range of input voltage ranges, Equation 3 could be utilized to find the expected gain, to which Equation 2 could be utilized to develop an inverting gain amplifier that 3 V to 10 C and 0.3V to 20 C. Demonstrated in Figure 3, is a schematic of the thermistor circuit that was designed.

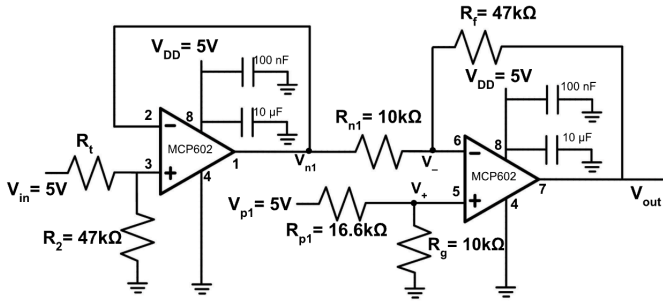


Figure 3: Thermistor Sensor Resistance-to-Voltage Interface Circuit, includes Unity Gain Buffer

With the circuit designed and tested, the sensor had to be calibrated to the temperature ranges of the design. This meant individually measuring different temperatures of water and the output voltage the thermistor outputs when connected to the circuit breadboard. This would result in a calibration curve that looks as such:

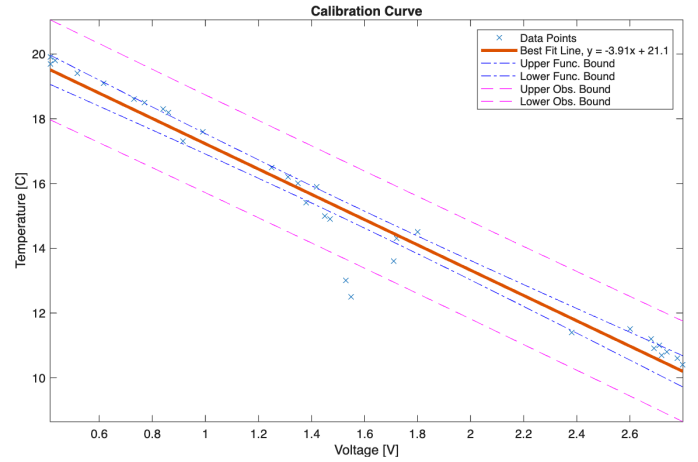


Figure 4: Calibration curve of Temperature (C) vs. Voltage (V)

As conveyed in Figure 4, a negative relationship is evident between temperature and voltage. To develop the calibration curve lab experiments were performed where a variety of temperatures were measured between the ranges 10-20 C by a thermometer to get the true value, and then measured again by the thermistor to get the output voltage after the inverting gain amplifier was applied. To increase accuracy the volume of data was collected was extremely important. Furthermore by parsing the data equally between 10-20, the best fit line would not be disproportionately influenced by a few data points.

C. UV Sensor

Finally the last sensor implemented on the diving robot was UV. The UV sensor chosen was the UV-V light sensor (GUVA-S12SD) chosen for its accuracy, cost, and built in calibration [4]. The sensor implements a voltage-to-voltage signal interface that is able to measure UV light intensity. The datasheet gives a UV to output relationship, given as such in Equation 7:

$$V_{out} = UV * 0.1 \quad (7)$$

Thus for example at a measured UV of 5, the output voltage of the sensor is 0.5 V demonstrated with Equation 7. Thus to develop a circuit that would fit the range of the teensy, it was important to know the possible input values. Since Dana Point varied UV index, a large range was chosen: 0-8 possible UV. Thus with the range settled, an inverting gain amplifier was utilized that mapped 0.3 V to 8 UV and 3V to 0 UV through leveraging Equation 3 to find the appropriate gain and offset and Equation 2 to map the output values. This was done through leveraging Equation 7 to get the possible input Voltages needed to develop a satisfactory gain for the circuit. Demonstrated in Figure 5, is a schematic of the UV circuit that was designed.

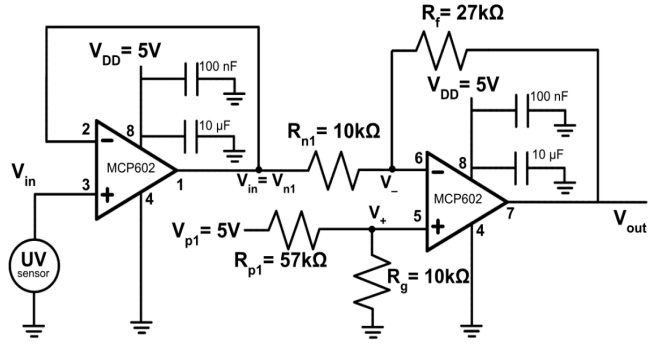


Figure 5: UV Sensor Voltage-to-Voltage Interface Circuit, includes Unity Gain Buffer

With the circuit developed, it was important to ensure that the circuit was accurately measuring the UV. Since a calibration curve wasn't necessary since the sensor itself converted UV to voltage, field experiments were performed to evaluate the accuracy of the UV sensor. Demonstrated in Figure 6 is the field experiment performed to ensure the accuracy of the UV sensor.

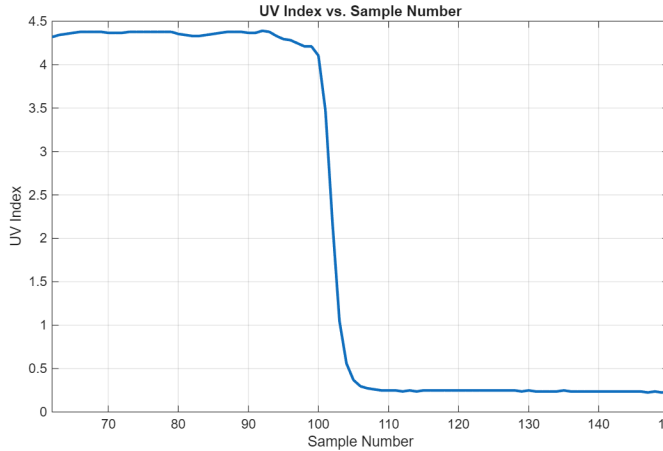


Figure 6: Field experiment to test accuracy of UV sensor

As conveyed in Figure 6, a relatively constant UV index of 4 was measured, which matched the UV reading according to the Weather Channel that day. However a drop occurs suddenly, which is due to sudden coverage of the UV sensor. This drop matches understanding of how a UV sensor works. Since the hand covered the majority of the light the UV could read, the UV sensor would assume that there would be very little UV entering the atmosphere. Thus the field testing conveys that the UV matches the expected results.

The standard 10Hz sampling rate was enough in producing results that accounted for the dynamic changes in the system. As it will be conveyed in the results section, the temperature sensor was most affected by noise due to motors not a slow sampling rate, and UV data collected was able to convey all the characteristics of an exponentially decaying function. Pressure was most affected by the environment, thus a change in sampling rate wouldn't affect the result.

III. ROBOT DESIGN

The robot was designed with 3 factors in mind: stability, neutral buoyancy, as well as spacing. To effectively travel downwards a rigid square sized robot allowed for minimal axial tilt, due to symmetry about the vertical axis. Next, since an important trait of the robot was that it had to be neutrally buoyant, the box that held all electronics was placed significantly above the ballasts in order to ensure that the center of buoyancy was above the center of mass. Equation 8 and 9 were leveraged to find the approximate locations of center of mass and center of buoyancy:

$$COM = (\int x dm)/M \quad (8)$$

$$COB = (\int x dB)/FB \quad (9)$$

The box being majority composed of air effectively raised the buoyancy of the robot, meaning that ballasts added had to account for this change. To find the approximate mass of ballasts needed Equation 10 was utilized. Here F_b refers to the buoyancy force acting on the robot, g the gravitational acceleration, and m_{robot} the mass of the robot including frame and all electronic components.

$$m_{ballasts} = \frac{F_b}{g} - m_{robot} \quad (10)$$

Finally due to electromagnetic interference with electronics, motors had to be placed at least 30 cm away from all electronics to reduce noise. This meant having a sizable gap between the box and the motors. This meant having a sizable gap between the box and the motors. Finally motor mount accessories were designed and 3D printed in order to hold the motors equally distance apart and ensure stability of the motors. These motors also assisted in making sure that the motors weren't directly aligned with the box, so that when the motors were on the water being pushed from the motors weren't elevating the box making the robot unable to move downwards. Refer to Figure 7 of a mechanical diagram of the robot and the robot deployed at Dana Point.

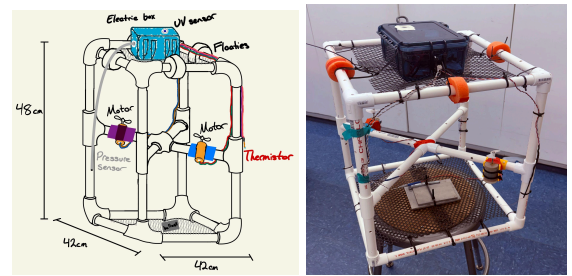


Figure 7: Mechanical diagram of robot and the robot that was deployed at Dana Point

As seen in Figure 7, to ensure no UV was being absorbed or reflected by the electrical box, a quartz window was applied to ensure the UV sensor would be able to detect incoming light.

IV. DEPLOYMENT DETAILS

During the day of Dana Point, there was a large overcast in the morning which reduced the amount of UV that was passed through and captured by the UV sensor. The robot was deployed at the dock, where waterproofing and neutral buoyancy were tested. Since, the density of saltwater is different compared to fresh water, the amount of ballasts needed changed to ensure neutral buoyancy. With all safety protocols accounted for such as ensuring a tight tether and members were safely positioned to deploy, ground truth measurements were taken. The temperature at Dana Point Pier was approximately 18.7 C, UV rating was approximately 2, and the atmosphere pressure was approximately 101.3kpa at 8:30 am. Once the robot was neutrally buoyant, the robot deployment was ready leading to data being collected. To deploy the robot, an external laptop was utilized to upload all the code needed to run the robot, and an SD card was used to log all the data into .log and .txt files that were later analyzed through matlab scripts. The robot was deployed at the OC Country Pier & Dock, as seen in Figure 8.

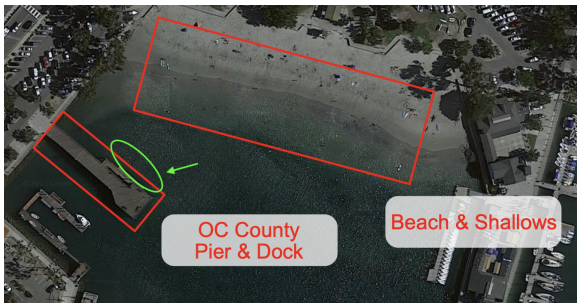


Figure 8: Top-down view of the harbor of Dana Point. The green circle highlights the dock where AUV deployment occurred.

V. RESULTS

After deployment at Dana Point Pier, the data was collected and stored in an sd card which was later used to analyze the respective .log and .txt files through a matlab logreader file. What was generated were the following plots. To begin, the first plot generated was the robots position as a function of time, as seen in Figure 9.

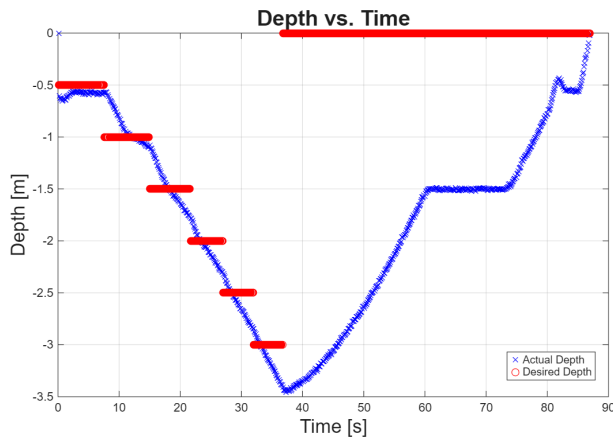


Figure 9: Depth vs. Depth Desired showing the control feedback of the robot adjusting its position to reach each waypoint

As seen in Figure 9, a plot is generated of the robot's position as a function of time. What can be first analyzed with this plot is how the robot began at a start position near 0.5 m. Since the pressure tube was placed near the bottom of the robot, and since the robot began neutrally buoyant in the water, the robot's position, marked in blue, wasn't able to begin at 0 depth. Next, from times 10 to about 34 seconds, it can be seen that the robot's control effort was achieved and the robot was able to navigate to each waypoint, desired position marked as red, and stay there for a relatively few seconds. In the designed program, to mitigate as much noise as possible which will be apparent in the temperature data, a motor stop was implemented in order to reduce as much noise as possible. Since salt water is extremely conductive, the electromagnetic interference becomes stronger leading to more noise. A stop of 4 seconds was implemented at each waypoint, however as seen by figure 9 the robot was only able to stay stationary for about 2 seconds. Through inference this conveys that the increased pressure at different depths led to the robot not being neutrally buoyant like it was on the surface. Thus while the robot did achieve each of its waypoints, at its last waypoint the robot began sinking making it unable to resurface back to the top of the water. The dip from 3 meters to 3.5 meters is most noticeable as the robot sinking, and after 38 seconds it can be seen as the robot being pulled back to the surface.

With the robots position data, a plot of UV was generated as a function of depth. Refer to Figure 10.

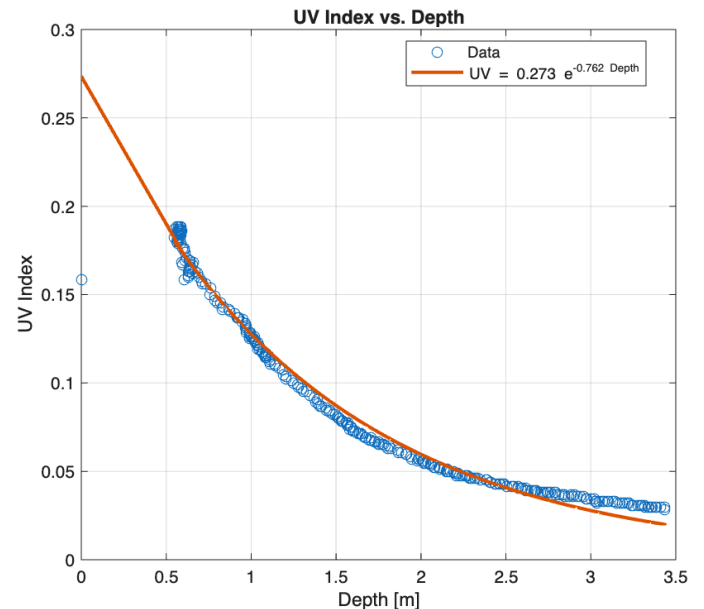


Figure 10: Figure of UV index vs. Depth, where UV decrease as a function of depth

As seen by Figure 10, UV decays exponentially which matches theoretical expectations [8]. This decrease is expected when viewing this model through Beer-Lambert's Law in

Equation 1. Where the data produced can be seen in Equation 11:

$$UV = 0.273 * \exp(-0.762 * Depth) \quad (11)$$

This means that as depth increases that UV is expected to decay exponentially. Furthermore, when analyzing the data present it can be seen that the decrease of UV per 1 meter depth is approximately, $e^{-0.762} = 46.2\%$. With a starting UV of about 2 which was measured as the ground truth measurement at Dana Point Pier at 8:30 am, UV decreased from 0.189 to 0.03, a decrease factor of 84% from 0 to 3 meters. This practically means that at a depth of 3 meters, there was practically no UV to be collected.

The last data that was collected at the deployment in Dana Point was temperature. Refer to Figure 11.

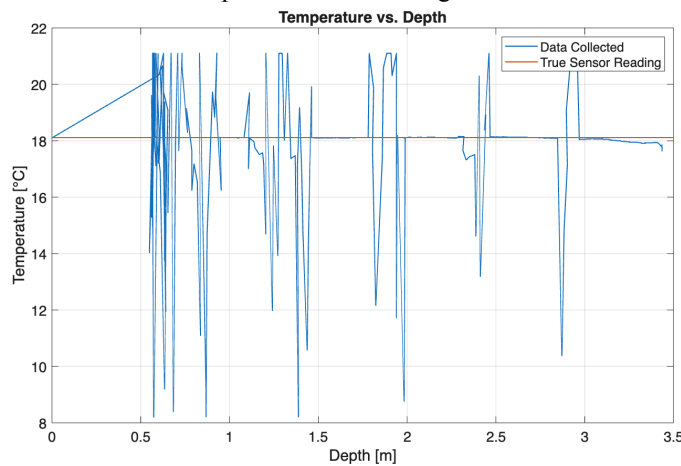


Figure 11: Temperature vs. Depth where temperature stays relatively constant at different depths that were measured

As seen with Figure 11, the plot of temperature with respect to depth demonstrates extremely noisy data. As mentioned previously, in an attempt to reduce the noise inflicted by the motors running in salty water, a stop was implemented which would stop the motors running at each waypoint. These gaps can be seen as the points in the plot where there aren't large vertical spikes. These values of temperature can be represented as the "True Sensor Reading" where noise of the motor isn't affecting the output temperature. Through analysis of each gap, the temperature values can be analyzed at each depth to create a representative reading of what the thermistor would have read if there wasn't any noise. In the case of the data collected it would be a constant 18.1 C, conveying that temperature stayed constant from depths 0 to 3 meters. The ground measurement collected at Dana Point Dock was measured by a digital thermometer to be 18.7 C. Finally the difference of 0.6 C, can be explained by a few factors. First the thermistor is a relatively cheap part that

has a margin of uncertainty, and the same holds true with the thermometer.

VI. CONCLUSION

Overall the results produced by the E80 AUV are extremely satisfactory and representative of the real world phenomena. To begin, as demonstrated in Depth vs. time the robot was able to accurately dive to each waypoint as shown by the overlap in depth actual and depth desired. However this wasn't without its imperfections. As the robot would descend and as depth increased the robot no longer would stay neutrally buoyant leading to a sinking effect making it unable to resurface itself back up to the surface of the water. Next when analyzing data for UV, an exponentially decay relationship was observed between UV and Depth, which matches theoretical expectations. This behavior conveys that since aquatic organisms are adapted to specific levels of UV exposure at different depths then climate change effectively changing the quality of water and altering the amount of UV entering can disrupt many of the organisms from what they are used to. Finally although aquatic life can be sensitive to small temperature changes, the measured temperature stayed relatively constant from 0-3 m. This suggests that temperature was not strongly affected by depth in the shallow range tested.

Although the data collected for temperature demonstrated signs of being constant, extreme fluctuations made data analysis extremely difficult. The thermistor being near the motors, produced a lot of noise due to their electromagnetic interference. To combat this a motor stop was implemented at each waypoint to reduce noise and create concrete data markers of what the temperature was at each waypoint. For future implementations, the thermistor would be placed at a reasonable distance away from the motors to ensure no interference would be detected. Furthermore, given the opportunity to retest at Dana point, another field measurement at Dana Point could be useful to compare temperatures and UV data at different locations in the harbor, potentially in the middle of the beach. Furthermore, to get even more interesting data, testing the robot at the same locations but at different times of the day would also produce interesting temperature and UV results

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REFERENCES

- [1] "E80 Project: Field Deployment of an Autonomous Robot – HMC E80." Accessed: May 07, 2026. [Online]. Available: <https://hmc-e80.github.io/project/overview/>
- [2] "NXFT15WB473FA2B150|NTC for Temperature Sensor|NTC Thermistors|Thermistors|Murata Manufacturing Co., Ltd." Accessed: May 07, 2026.

[Online]. Available:
<https://pim.murata.com/en-global/pim/details/?partNum=NXFT15WB473FA2B150>

- [3] Freescale Semiconductor, Inc., “Pressure Sensor”, “MPX5700 Series: Piezoresistive Transducers,” Datasheet, [Revised 10/10/2012], Accessed: May 07, 2026. [Online]. Available:
<https://www.nxp.com/docs/en/data-sheet/MPX5700.pdf>
- [4] Adafruit Industries LLC, “UV Sensor”, “GUVA-S12D” Datasheet, Accessed: May 07, 2026. [Online]. Available:
<https://cdn-shop.adafruit.com/datasheets/1918guva.pdf>
- [5] D.-P. Häder, E. W. Helbling, C. E. Williamson, and R. C. Worrest, “Effects of UV radiation on aquatic ecosystems and interactions with climate change,” *Photochem. Photobiol. Sci. Off. J. Eur. Photochem. Assoc. Eur. Soc. Photobiol.*, vol. 10, no. 2, pp. 242–260, Feb. 2011, doi: 10.1039/c0pp90036b.
- [6] P. A. k *et al.*, “Impact of climate change and anthropogenic activities on aquatic ecosystem – A review,” *Environ. Res.*, vol. 238, p. 117233, Dec. 2023, doi: 10.1016/j.envres.2023.117233.
- [7] “How rising water temperatures impact marine and aquatic life,” IFAW. Accessed: May 07, 2026. [Online]. Available:
<https://www.ifaw.org/international/journal/rising-water-temperatures-impact-marine-aquatic-life>
- [8] C. J. Weiskerger, M. D. Rowe, C. A. Stow, D. Stuart, and T. Johengen, “Application of the Beer–Lambert Model to Attenuation of Photosynthetically Active Radiation in a Shallow, Eutrophic Lake”, doi: 10.1029/2018WR023024.
- [9] “Dana Point (CA) sea water temperature in April.” Accessed: May 07, 2026. [Online]. Available:
<https://seatemperature.info/april/dana-point-water-temperature.html>